

Laporan Praktikum (WEB SERVER)

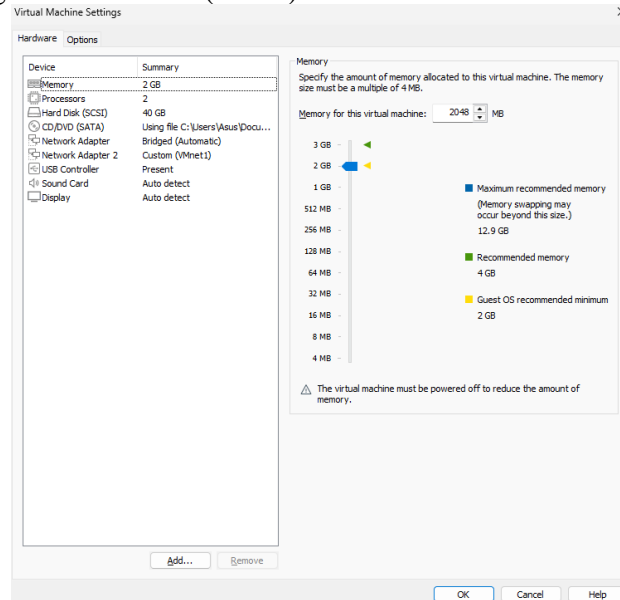


NAMA:	Bona Maruli Panjaitan
NIM:	43324035
PRODI:	D3 Teknologi Komputer

**INSTITUT TEKNOLOGI DEL
FAKULTAS VOKASI**

Konfigurasi DHCP:

1. Pastikan VM mempunyai 2 NIC/Network Adapter. Adapter pertama menggunakan bridge, ini digunakan untuk mengkoneksikan ubuntu server ke internet, sehingga ip pada Network adapter ini akan secara otomatis diberikan (Dynamic). Adapter kedua menggunakan custom (contohnya VMnet1) dan ip pada adapter ini akan di setting secara manual (Static).



2. Cek ip nya, harus mendapatkan IP untuk NIC nya. Jika belum ada maka perlu di setting ip secara static. Pergi ke direktori /etc/netplan. Lalu edit file .yaml yang ada pada direktori itu.

```
GNU nano 7.2 /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens34:
      dhcp4: no
      addresses:
        - 192.168.12.1/24
      nameservers:
        addresses: [192.168.12.1, 8.8.8.8]
```

3. Kemudian terapkan konfigurasi tersebut dengan menegtikkan perintah berikut.

```
sudo netplan apply
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo netplan apply
bonapanjaitan556@bonapanjaitan556:~$
```

4. Cek ip address nya kembali dan pastikan sudah ada.

```
ip a
```

```
bonapanjaitan556@bonapanjaitan556:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:73:d7:c8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.54.232.71/24 metric 100 brd 10.54.232.255 scope global dynamic ens33
        valid_lft 3544sec preferred_lft 3544sec
    inet6 fe80::20c:29ff:fe73:d7c8/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:73:d7:d2 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.12.1/24 brd 192.168.12.255 scope global ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe73:d7d2/64 scope link
        valid_lft forever preferred_lft forever
bonapanjaitan556@bonapanjaitan556:~$ _
```

5. Install DHCP Server menggunakan perintah

```
sudo apt install isc-dhcp-server
```

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bonapanjaitan556@bonapanjaitan556:~$ _
```

6. Konfigruasi isc-dhcp-server. Untuk pengkonfigurasian dilakukan pada file /etc/dhcp/dhcpd.conf. Cari pada baris # A slightly different configuration for an internal subnet. Kemudian hapus karakter '#' pada konfigurasi dibawahnya. Menjadi seperti ini.

```
sudo nano /etc/dhcp/dhcpd.conf
```

```
# A slightly different configuration for an internal subnet.
subnet 192.168.12.0 netmask 255.255.255.0 {
    range 192.168.12.10 192.168.12.100;
    option domain-name-servers ns1.internal.example.org;
    option domain-name "internal.example.org";
    option subnet-mask 255.255.255.0;
    option routers 192.168.12.1;
    option broadcast-address 192.168.12.255;
    default-lease-time 600;
    max-lease-time 7200;
}
```

7. Tentukan interface mana yang akan menjalankan DHCP dengan mengkonfigurasi pada file /etc/default/isc-

dhcp-server.

```
sudo nano /etc/default/isc-dhcp-server
```

```
GNU nano 7.2 /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

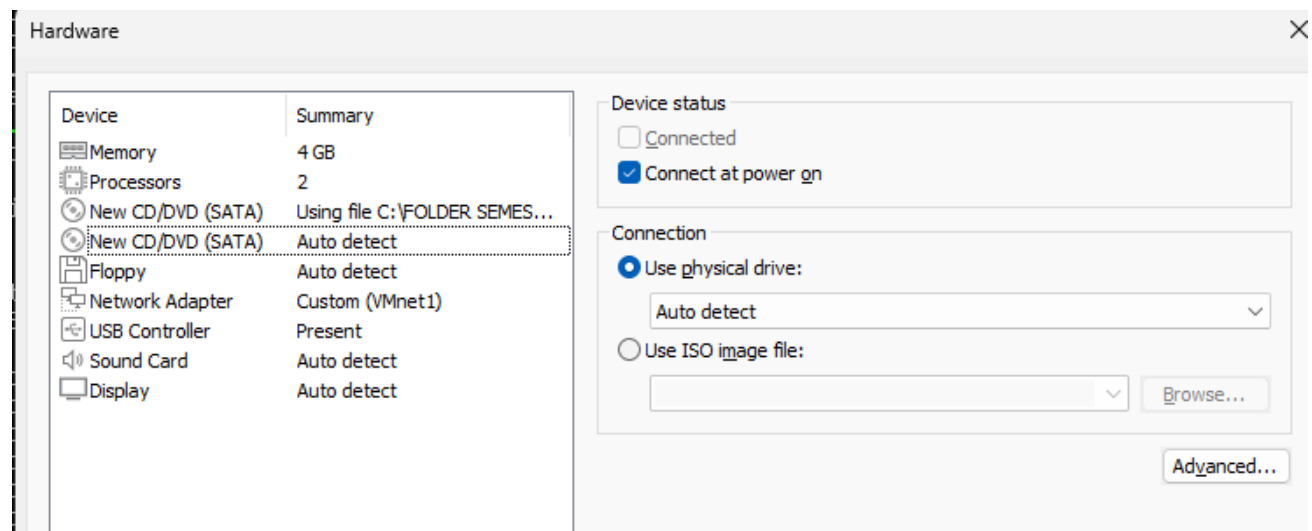
# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens34"
INTERFACESv6=""
```

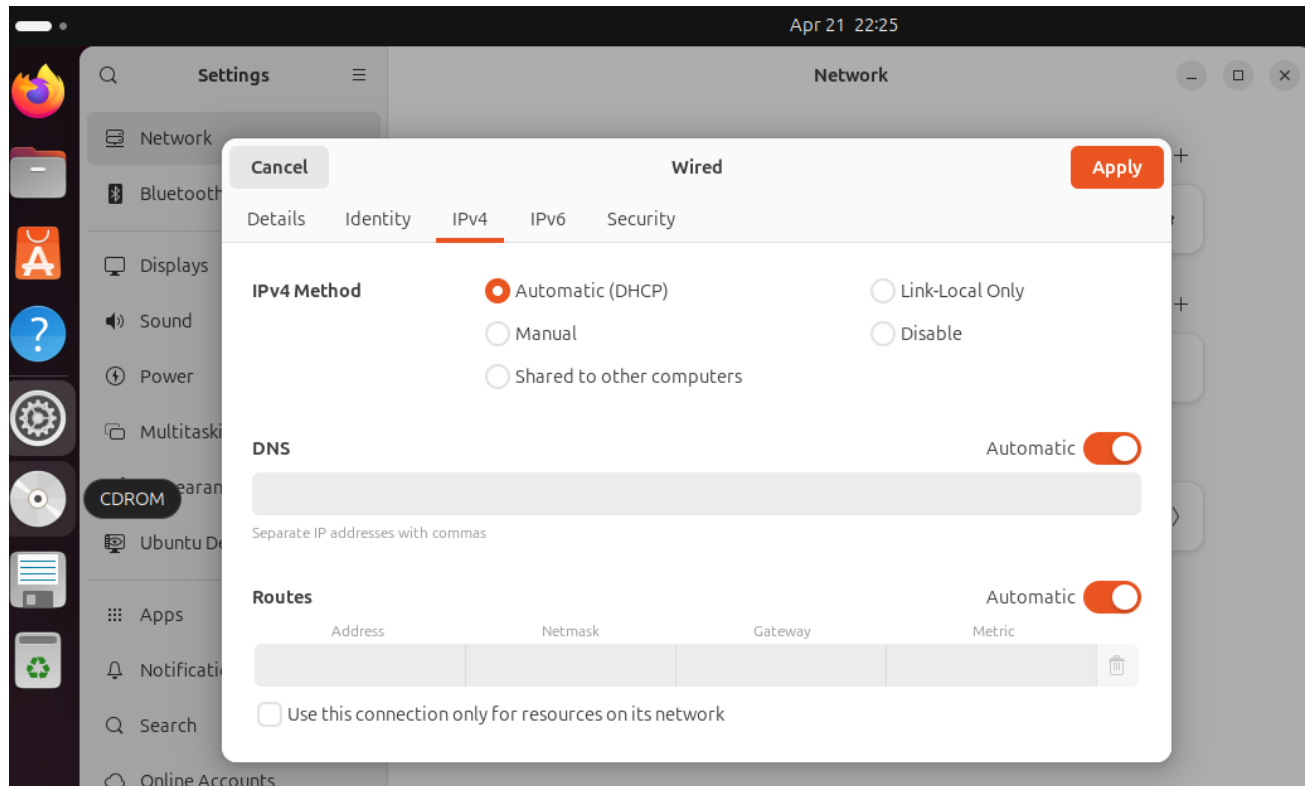
8. Setelah konfigurasinya selesai restart service DHCP nya untuk menerapkan perubahan konfigurasi yang sudah dilakukan.

```
sudo systemctl restart isc-dhcp-server.service
sudo systemctl status isc-dhcp-server
```

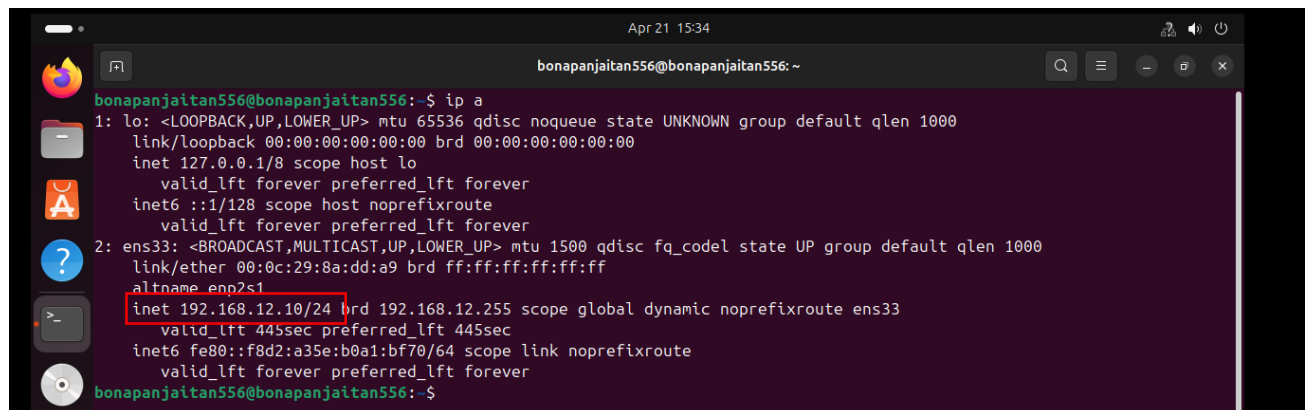
9. Sebelum melakukan pengujian kita harus melakukan pengaturan pada vm. Pengujian dilakukan dengan menggunakan vm lainnya (disini menggunakan ubuntu desktop). Untuk settingan network adaptornya menggunakan custom dan pastikan sama dengan adapter kedua pada ubuntu server. Jadi jika adapter kedua pada ubuntu server menggunakan custom: VMnet1 maka pada ubuntu desktop juga wajib menggunakan custom: VMnet1. Ini artinya NIC 2 pada ubuntu server dengan NIC pada ubuntu desktop terhubung secara fisik.



10. Untuk melakukan pengujian, pastikan pengaturan ipv4 nya menggunakan Automatic bukan Manual, Shared to other computers, Link-Local Only, dan Disable.



11. Buka terminal pada ubuntu desktop dan ketikkan perintah ip a untuk melihat ip addressnya. Pastikan untuk ip networknya sama dengan nic network pada interface 2 pada ubuntu server.



Konfigurasi NAT:

1. Aktifkan ip forwarding dengan cara edit file /etc/sysctl.conf, kemudian cari baris #net.ipv4.ip_forward=1, lalu hapus “#” untuk mengaktifkan nya. Lalu ketikkan perintah sudo sysctl -p untuk mengaktifkan/menjalankan.

```
sudo nano /etc/sysctl.conf
sudo sysctl -p
```

```
# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

2. Sekarang kita perlu mengkonfigurasi ip tablesnya dengan mengetikkan perintah berikut.

```
sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
```

3. Izinkan forward.

```
sudo iptables -A FORWARD -i ens33 -o ens34 -m state --state
RELATED,ESTABLISHED -j ACCEPT
```

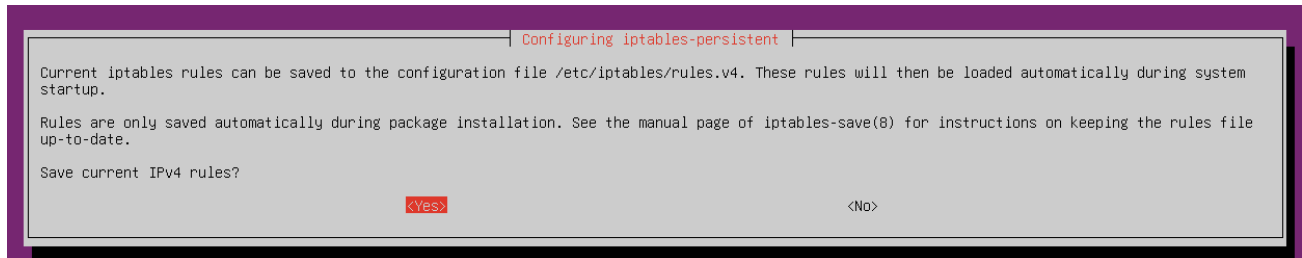
```
sudo iptables -A FORWARD -i ens34 -o ens33 -j ACCEPT
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo iptables -A FORWARD -i ens33 -o ens34 -m state --state RELATED,ESTABLISHED -j ACCEPT
bonapanjaitan556@bonapanjaitan556:~$ sudo iptables -A FORWARD -i ens34 -o ens33 -j ACCEPT
bonapanjaitan556@bonapanjaitan556:~$ _
```

4. Simpan iptables agar konfigurasi tidak hilang saat reboot. Install iptables-persistent.

```
sudo apt install iptables-persistent
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install iptables-persistent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  netfilter-persistent
The following packages will be REMOVED:
  ufw
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 1 to remove and 63 not upgraded.
Need to get 14.3 kB of archives.
After this operation, 780 kB disk space will be freed.
Do you want to continue? [Y/n] _
```



5. Kemudian simpan iptablesnya.

```
sudo netfilter-persistent save
```

6. Kemudian lakukan pengujian dengan mengetikkan perintah ping 8.8.8.8 pada ubuntu desktop (client) untuk menguji apakah konfigurasi NAT sudah berhasil atau belum, jika sudah hasil dari ping tersebut akan menghasilkan TTL.

```
bonapanjaitan556@bonapanjaitan556: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
Files  
bonapanjaitan556@bonapanjaitan556:~$ ping 8.8.4.4 -c 4  
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.  
64 bytes from 8.8.4.4: icmp_seq=1 ttl=113 time=63.5 ms  
64 bytes from 8.8.4.4: icmp_seq=2 ttl=113 time=153 ms  
64 bytes from 8.8.4.4: icmp_seq=3 ttl=113 time=75.6 ms  
64 bytes from 8.8.4.4: icmp_seq=4 ttl=113 time=76.0 ms  
  
--- 8.8.4.4 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 63.463/92.101/153.361/35.724 ms  
bonapanjaitan556@bonapanjaitan556:~$ ping google.com  
PING google.com (172.253.118.139) 56(84) bytes of data.  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=1 ttl=103 time=60.5 ms  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=2 ttl=103 time=65.6 ms  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=3 ttl=103 time=103 ms  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=4 ttl=103 time=73.9 ms  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=5 ttl=103 time=57.3 ms  
64 bytes from sl-in-f139.1e100.net (172.253.118.139): icmp_seq=6 ttl=103 time=48.9 ms  
^C  
--- google.com ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5009ms  
rtt min/avg/max/mdev = 48.906/68.166/102.726/17.229 ms  
bonapanjaitan556@bonapanjaitan556:~$
```

7. IP TABLES

```
sudo iptables -t nat -L -v -n
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo iptables -t nat -L -v -n
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination

Chain POSTROUTING (policy ACCEPT 5 packets, 401 bytes)
 pkts bytes target    prot opt in     out     source            destination
  13   934 MASQUERADE 0    --  *      ens33  0.0.0.0/0         0.0.0.0/0
bonapanjaitan556@bonapanjaitan556:~$ _
```

8. Pada konfigurasi NAT ini sudah terbukti bahwa, client sudah bisa ping ke server google yaitu 8.8.8.8, namun ketika ping ke google.com hasilnya adalah temporary failure in name resolution. Itu bukan berarti konfigurasi NAT tidak berhasil, namun karena itu perlu konfigurasi service yang lain yaitu DNS. Dengan client dapat ping ke server google (8.8.8.8) itu sudah cukup membuktikan bahwa client dapat terkoneksi ke internet.

Konfigurasi DNS Pada Ubuntu Server

1. Update dan upgrade ubuntu server anda sebelum menginstall dan memulai konfigurasi DNS server pada ubuntu server anda dengan command

```
sudo apt update && sudo apt upgrade -y
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
0% [Working]
```

2. Setelah selesai update dan upgrade ubuntu server lakukan instalasi bind9 yang merupakan software DNS server paling umum digunakan di Linux dengan command sudo apt install bind9. Tampilan dibawah Adalah tampilan ketika sudah berhasil install bind9.

```
Sudo apt install bind9
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install bind9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-utils dns-root-data
Suggested packages:
  bind-doc ufw
The following NEW packages will be installed:
  bind9 bind9-utils dns-root-data
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 423 kB of archives.
After this operation, 1,631 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

3. Cek status bind9 dengan command berikut :

```
sudo systemctl start bind9
```

```
sudo systemctl enable bind9
```

```
sudo systemctl status bind9
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl start bind9
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl enable bind9
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl status bind9
• named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 13:48:59 UTC; 2min 23s ago
     Docs: man:named(8)
   Main PID: 11288 (named)
    Status: "running"
   Tasks: 8 (limit: 2205)
  Memory: 23.9M (peak: 24.3M)
     CPU: 51ms
   CGroup: /system.slice/named.service
           └─11288 /usr/sbin/named -f -u bind

Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './DNSKEY/IN': 2001:503:c27::2:30#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::f#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './DNSKEY/IN': 2001:503:ba3e::2:30#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './NS/IN': 2001:503:ba3e::2:30#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Apr 21 13:48:59 bonapanjaitan556 named[11288]: managed-keys-zone: Initializing automatic trust anchor management for zone '.': DNSKEY ID 20326 is now trusted,
Apr 21 13:48:59 bonapanjaitan556 named[11288]: managed-keys-zone: Initializing automatic trust anchor management for zone '.': DNSKEY ID 38696 is now trusted,
lines 1-22/22 (END)
```

4. Masuk ke file konfigurasi DNS dengan command `sudo nano /etc/bind/named.conf.options` yang merupakan direktori konfigurasi DNS server. Lakukan konfigurasi pada file tersebut
IP yang digunakan Adalah IP DNS yang sudah kita subnetting di DHCP server

```
sudo nano /etc/bind/named.conf.options
```

```
GNU nano 7.2 /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    // Mengizinkan query dari mana saja
    allow-query { any; };

    recursion yes;

    // Izinkan rekursi dari jaringan lokal ANDA
    allow-recursion { localhost; 192.168.12.0/24; };

    // Dengarkan hanya dari IP Server Anda dan loopback
    listen-on { 127.0.0.1; 192.168.12.1; };_
    listen-on-v6 { any; };
};
```

5. Akses file konfigurasi bind untuk mendaftarkan zona yaitu forward dan reverse dengan command

```
sudo nano /etc/bind/named.conf.local
```

Forward : fungsinya untuk menerjemahkan domain name à IP

Reverse : fungsinya untuk menerjemahkan IP à domain name

```
GNU nano 7.2 /etc/bind/named.conf.local *
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "bona.local" {
    type master;
    file "/etc/bind/zones/forward.bona";
    allow-update { none; };
};

zone "12.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/reverse.bona";
    allow-update { none; };
};
```

6. Sekarang buatlah directory untuk membuat zona file tadi disini saya membuat nya di dalam /etc/bind/zones dengan command

```
sudo mkdir /etc/bind/zones
```

7. Selanjutnya buatkan file zone forward tadi dengan command ketikan berikut :

```
sudo nano /etc/bind/zones/forward.bona
```

```
GNU nano 7.2 /etc/bind/zones/forward.bona *
$TTL      604800

@          IN      SOA      ns1.bona.local. admin.bona.local. (
                                2024052201      ; Serial
                                3600              ; Refresh
                                1800              ; Retry
                                604800            ; Expire
                                86400 )          ; Negative Cache TTL
;

@          IN      NS       ns1.bona.local.
@          IN      A        192.168.12.1
ns1        IN      A        192.168.12.1
server     IN      A        192.168.12.1
www        IN      A        192.168.12.1
```

8. Selanjutnya buatkan file zone reverse tadi dengan command ketikan berikut :

```
sudo nano /etc/bind/zones/reverse.bona
```

```
GNU nano 7.2 /etc/bind/zones/reverse.bona *
$TTL      604800

@          IN      SOA      ns1.bona.local. admin.bona.local. (
                                2024052201      ; Serial
                                3600              ; Refresh
                                1800              ; Retry
                                604800            ; Expire
                                86400 )          ; Negative Cache TTL
;

@          IN      NS       ns1.bona.local.
1          IN      PTR      ns1.bona.local.
1          IN      PTR      server.bona.local.
```

9. Kemudian lakukan pengecekan sysntex konfigurasi dengan command berikut :

sudo named-checkconf

sudo named-checkzone bona.local /etc/bind/zones/forward.bona

sudo named-checkzone 12.168.192.in-addr.arpa /etc/bind/zones/reverse.bona

```
bonapanjaitan556@bonapanjaitan556:~$ sudo named-checkconf
bonapanjaitan556@bonapanjaitan556:~$ sudo named-checkzone bona.local /etc/bind/zones/forward.bona
zone bona.local/IN: loaded serial 2024052201
OK
bonapanjaitan556@bonapanjaitan556:~$ sudo named-checkzone 12.168.192.in-addr.arpa /etc/bind/zones/reverse.bona
zone 12.168.192.in-addr.arpa/IN: loaded serial 2024052201
OK
bonapanjaitan556@bonapanjaitan556:~$ _
```

10. Selanjutnya restart dan aktifkan bind9 agar semua konfigurasi kita tadi terbaca kemudian cek Kembali status bind9

```
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl restart bind9

bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl enable bind9

Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl status bind9

• named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 14:26:15 UTC; 18s ago
     Docs: man:named(8)
   Main PID: 11504 (named)
    Status: "running"
     Tasks: 6 (limit: 2205)
  Memory: 22.9M (peak: 23.3M)
     CPU: 44ms
   CGroup: /system.slice/named.service
           └─11504 /usr/sbin/named -f -u bind

Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:2d::d#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:1b8:10::b#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: network unreachable resolving './NS/IN': 2001:500:a8::e#53
Apr 21 14:26:15 bonapanjaitan556 named[11504]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Apr 21 14:26:15 bonapanjaitan556 named[11504]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
bonapanjaitan556@bonapanjaitan556:~$
```

11. Selanjutnya buka port DNS di firewall dengan command berikut

Jika menggunakan file ubuntu server sebagai iso nya maka ketik command ini:

```
sudo apt update && sudo apt install ufw -y
```

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bonapanjaitan556@bonapanjaitan556:~$
```

```
sudo ufw allow 53/tcp
```

```
sudo ufw allow 53/udp
```

```
sudo ufw enable
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo ufw allow 53/tcp
```

```
Rules updated
```

```
Rules updated (v6)
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo ufw allow 53/udp
```

```
Rules updated
```

```
Rules updated (v6)
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo ufw enable
```

```
Firewall is active and enabled on system startup
```

```
bonapanjaitan556@bonapanjaitan556:~$
```

Port 53 adalah port standar DNS, perlu dibuka untuk TCP dan UDP karena keduanya digunakan.

12. Pengujian dari server

1. Command nslookup domain_anda

```
bonapanjaitan556@bonapanjaitan556:~$ nslookup www.bona.local
```

```
Server:      192.168.12.1
```

```
Address:     192.168.12.1#53
```

```
Name:   www.bona.local
```

```
Address: 192.168.12.1
```

```
bonapanjaitan556@bonapanjaitan556:~$
```

2. Command dig domain_anda

```
bonapanjaitan556@bonapanjaitan556:~$ dig www.bona.local
```

```
; <<> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<> www.bona.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2294
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
```

```
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: fbe8ba7fec35bf570100000069e82b49d4ca7830c9222b5c (good)
;; QUESTION SECTION:
;www.bona.local.                IN      A
```

```
;; ANSWER SECTION:
www.bona.local.                604800  IN      A      192.168.12.1
```

```
;; Query time: 0 msec
;; SERVER: 192.168.12.1#53(192.168.12.1) (UDP)
;; WHEN: Wed Apr 22 01:58:33 UTC 2026
;; MSG SIZE rcvd: 87
```

```
bonapanjaitan556@bonapanjaitan556:~$
```

Konfigurasi Web Server pada Ubuntu Server

1. Lakukan update ubuntu dengan mengetik command:

```
sudo apt update
```

2. Lakukan penginstalan apache2 dengan mengetikkan command:

```
sudo apt install apache2
```

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bonapanjaitan556@bonapanjaitan556:~$ _
```

3. Install modul php

```
sudo apt install php libapache2-mod-php php-mysql
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install php libapache2-mod-php php-mysql
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
0 upgraded, 11 newly installed, 0 to remove and 5 not upgraded.
Need to get 5,051 kB of archives.
After this operation, 22.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

4. Install Mysql

```
sudo apt install mysql-server
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server-8.0
  mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinycat
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0
  mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 5 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

5. install ssh

```
sudo apt update
```

```
sudo apt install openssh-server -y
```

```
sudo systemctl status ssh
```

6. Izinkan SSH lewat Firewall Ubuntu:

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow ssh
```

```
sudo ufw reload
```

7. masuk ke windows (bukan VMWare) buka powerShell ketik perintah:

```
scp 14_Sistem_Data_Mobil.zip bonapanjaitan556@10.54.232.71:/home/bonapanjaitan556/
```

```
PS C:\xampp\htdocs> scp 14_Sistem_Data_Mobil.zip bonapanjaitan556@10.54.232.71:/home/bonapanjaitan556/
bonapanjaitan556@10.54.232.71's password:
14_Sistem_Data_Mobil.zip
PS C:\xampp\htdocs> |
```

100% 64MB 81.1MB/s 00:00

Note: untuk nama file .zip sesuaikan dengan nama file anda, untuk nama domain sesuaikan dengan nama vmware anda, ip sesuaikan dengan punya ip anda, folder sesuaikan dengan nama folder anda.

8. install aplikasi unzip di vmware

```
sudo apt update && sudo apt install unzip -y
```

9. Ekstrak file yang sudah masuk ke vmware

```
sudo unzip /home/bonapanjaitan556/14_Sistem_Data_Mobil.zip -d /var/www/html/
```

note: sesuaikan command nya dengan milik anda

8. Instalasi Tool Unzip pada Ubuntu Server

```
sudo apt update && sudo apt install unzip -y
```

9. Ekstraksi File Project Website

```
sudo unzip /home/bonapanjaitan556/14_Sistem_Data_Mobil.zip -d
/var/www/html/
```

10. Konfigurasi Database Management System (MySQL)

```
sudo mysql
```

Jalankan query berikut di dalam terminal MySQL:

```
CREATE DATABASE db_mobil;  
CREATE USER 'bona'@'localhost' IDENTIFIED BY 'password123';  
GRANT ALL PRIVILEGES ON db_mobil.* TO 'bona'@'localhost';  
FLUSH PRIVILEGES;  
EXIT;
```

11. Import Skema Database (.sql)

```
sudo mysql db_mobil < /home/bonapanjaitan556/"sistem-data-mobil (1).sql"
```

12. Konfigurasi Environment Aplikasi (Laravel)

```
cd /var/www/html
```

```
sudo nano .env
```

Isi konfigurasi:

```
DB_DATABASE=db_mobil  
DB_USERNAME=bona  
DB_PASSWORD=password123
```

13. Manajemen Izin Direktori dan Keamanan Aplikasi

```
sudo chown -R www-data:www-data /var/www/html  
sudo chmod -R 775 /var/www/html/storage  
sudo chmod -R 775 /var/www/html/bootstrap/cache  
php artisan key:generate
```


14. Konfigurasi Apache Virtual Host agar Akses DNS Tanpa "/public"

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

Hasil:

```
10.54.232.71 - PuTTY
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf
<VirtualHost *:80>
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/public

<Directory /var/www/html/public>
    AllowOverride All
    Require all granted
</Directory>

# Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Aktifkan modul rewrite dan restart layanan Apache:

```
sudo a2enmod rewrite
```

```
sudo systemctl restart apache2
```

15. Pengujian Akses dari Client (Ubuntu Desktop)

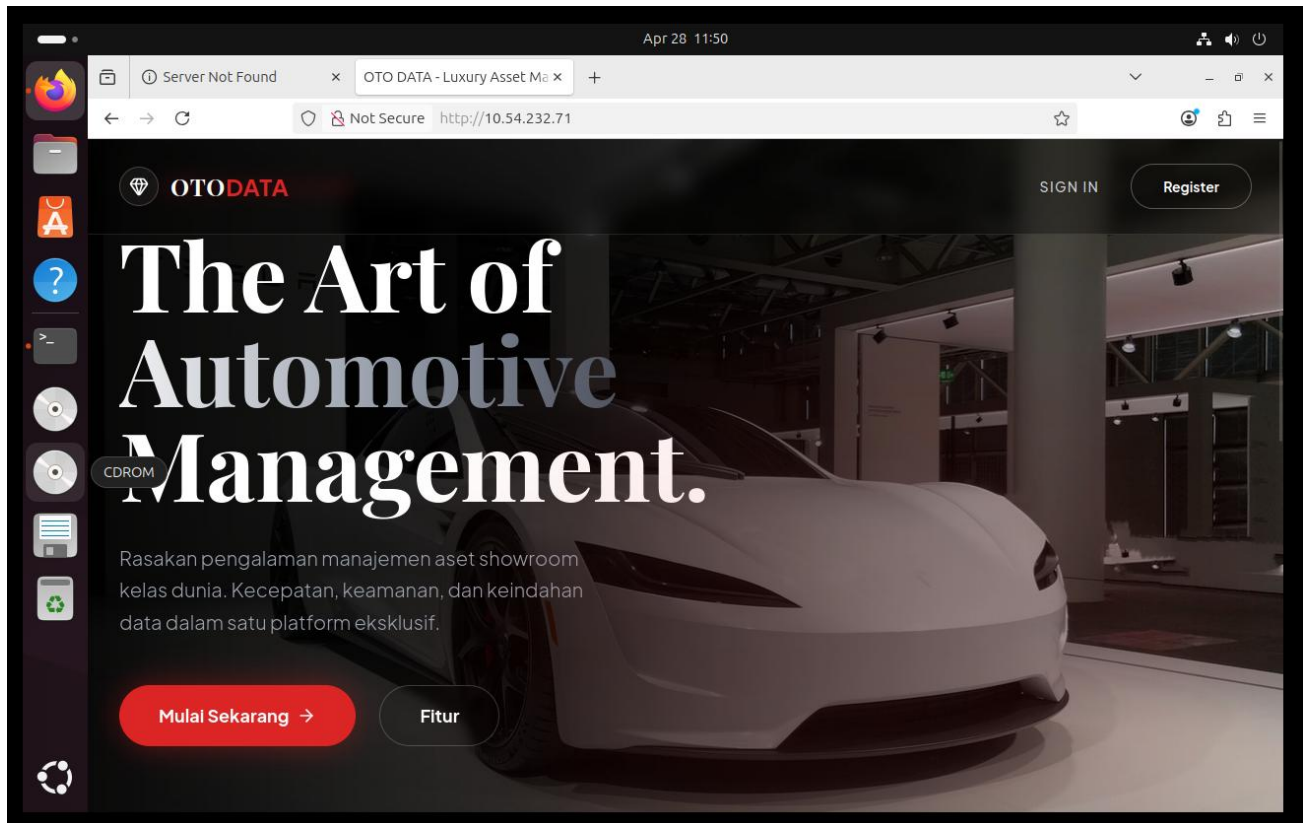
```
sudo nano /etc/hosts
```

```
bonapanjaitan556@bonapanjaitan556: ~
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 bonapanjaitan556

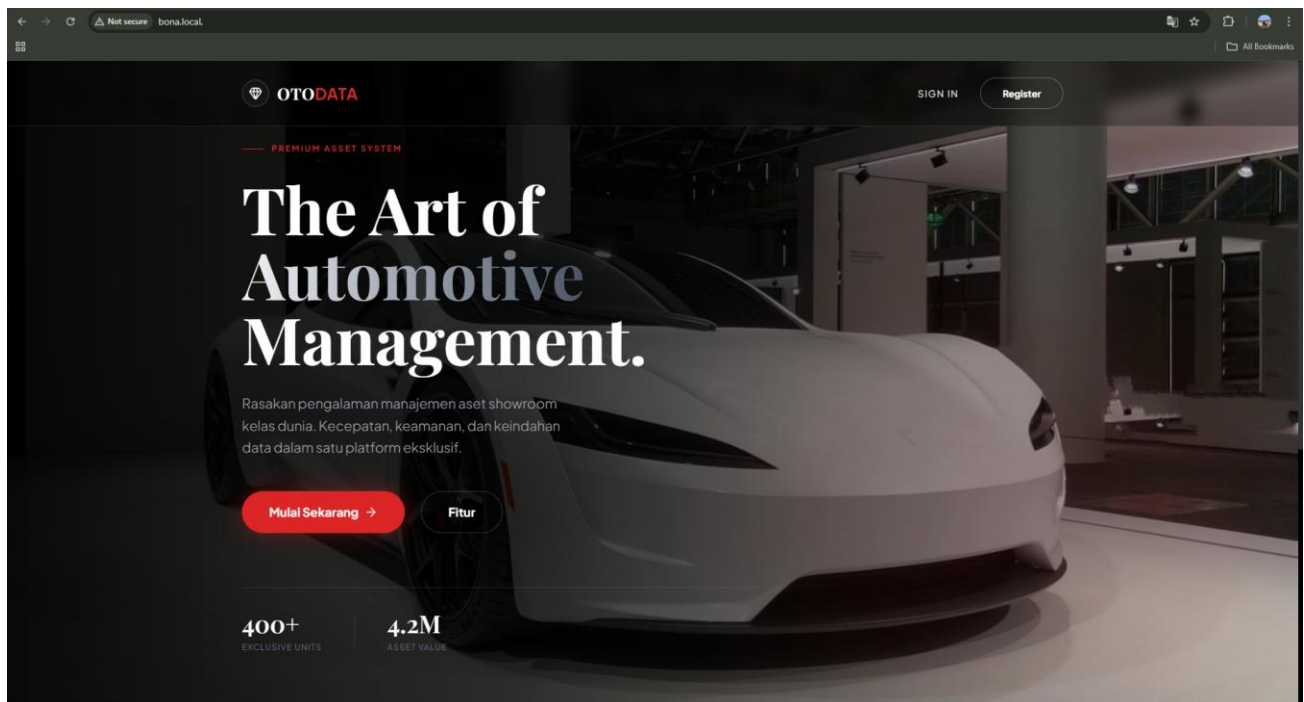
# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

192.168.12.1 bona.local
```

Buka *Web Browser* pada Ubuntu Desktop



Buka *Web Browser* pada Windows



Instalasi & Konfigurasi FTP Server di Ubuntu Server

1. Update vmware anda menggunakan command `sudo apt update` kemudian setelah selesai update unduh ubuntu server dengan command

```
sudo apt install vsftpd
```

Hasil:

10.54.232.71 - PuTTY

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install vsftpd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  vsftpd
0 upgraded, 1 newly installed, 0 to remove and 9 not upgraded.
Need to get 120 kB of archives.
After this operation, 312 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]
Fetched 120 kB in 2s (74.6 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database ... 90%
```

2. Aktifkan service ftp dengan command:

```
sudo systemctl start vsftpd
```

dan jika ingin otomatis hidup saat restart / booting machine gunakan command:
lalu lihat hasilnya dengan command status

```
sudo systemctl enable vsftpd
sudo systemctl status vsftpd
```

sudo sy

Hasi:

10.54.232.71 - PuTTY

```
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl start vsftpd
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-28 06:28:21 UTC; 2min 44s ago
     Main PID: 1787 (vsftpd)
       Tasks: 1 (limit: 2205)
      Memory: 712.0K (peak: 1.5M)
         CPU: 7ms
        CGroup: /system.slice/vsftpd.service
                └─1787 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 28 06:28:21 bonapanjaitan556 systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 28 06:28:21 bonapanjaitan556 systemd[1]: Started vsftpd.service - vsftpd FTP server.
bonapanjaitan556@bonapanjaitan556:~$
```

4. jika FTP server sudah berhasil berjalan selanjutnya masuk ke file konfigurasi FTP server directory berada di /etc/vsftpd.conf masuk dengan command:

```
sudo nano /etc/vsftpd.conf
```

```
10.54.232.71 - PuTTY
GNU nano 7.2 /etc/vsftpd.conf
# Example config file /etc/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
```

Cari baris seperti ini dan sesuaikan dengan konfigurasi pada kolom:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
chroot_local_user=YES
```

Bagian ini tambahkan di paling bawah pada konfigurasi !!

```
# Passive mode (penting agar client bisa konek dari luar)
pasv_enable=YES
pasv_min_port=10000
pasv_max_port=10100
```

5. Setelah selesai melakukan konfigurasi anda dapat restart FTP server dengan command sudo systemctl restart vsftpd dan cek status nya pastikan running jika gagal/eror bberarti ada kesalahan pada file konfigurasi.

```
Sudo systemctl restart vsftpd
Sudo systemctl status vsftpd
```

Hasil:

```
10.54.232.71 - PuTTY
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl restart vsftpd
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-28 06:48:52 UTC; 8s ago
     Process: 2158 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
    Main PID: 2160 (vsftpd)
      Tasks: 1 (limit: 2205)
     Memory: 716.0K (peak: 1.5M)
        CPU: 6ms
    CGroup: /system.slice/vsftpd.service
            └─2160 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 28 06:48:52 bonapanjaitan556 systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 28 06:48:52 bonapanjaitan556 systemd[1]: Started vsftpd.service - vsftpd FTP server.
bonapanjaitan556@bonapanjaitan556:~$
```

6. Selanjutnya kita akan membuat autentikasi untuk mengakses file di ftp server dengan command sudo adduser (namauser) selanjutnya akan masukan password dimana itu berguna untuk autentikasi saat ingin mengakses FTP server.

```
Sudo adduser userftp
```

Hasil:

```
10.54.232.71 - PuTTY
bonapanjaitan556@bonapanjaitan556:~$ sudo adduser userftp
info: Adding user `userftp' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `userftp' (1001) ...
info: Adding new user `userftp' (1001) with group `userftp (1001)' ...
info: Creating home directory `/home/userftp' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for userftp
Enter the new value, or press ENTER for the default
    Full Name []: userftp
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `userftp' to supplemental / extra groups `users' ...
info: Adding user `userftp' to group `users' ...
bonapanjaitan556@bonapanjaitan556:~$
```

7. Selanjutnya kita akan membuat folder yang akan diakses dan juga juga folder yang akan di upload dengan command sudo mkdir /home/ftpuser/ftpfile untuk file akses dan sudo mkdir /home/ftpuser/ftp/upload untuk file upload.

```
// Folder untuk Akses
mkdir /home/ftpuser/ftp
chmod 755 /home/ftpuser/ftp

//Folder untuk upload
mkdir /home/ftpuser/ftp/upload
chown ftpuser:ftpuser /home/ftpuser/ftp/upload
```

8. Atur permission wajib jika menggunakan root

```
sudo chown root:root /home/ftpuser
sudo chmod 755 /home/ftpuser
```

```
bonapanjaitan556@bonapanjaitan556:~$ sudo chown root:root /home/ftpuser
[sudo] password for bonapanjaitan556:
bonapanjaitan556@bonapanjaitan556:~$ sudo chmod 755 /home/ftpuser
bonapanjaitan556@bonapanjaitan556:~$
```

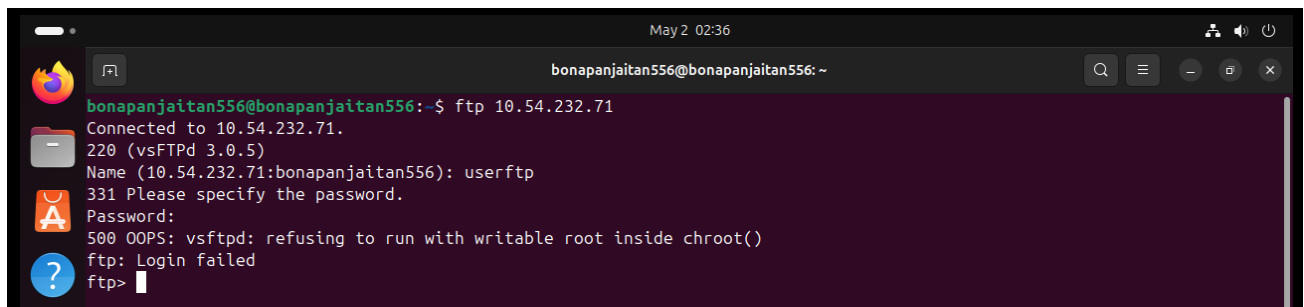
9. Buka firewall dengan port khusus untuk ftp yaitu port 21 dengan command

```
sudo ufw allow 21  
sudo ufw allow 10000:10100/tcp
```

Hasil:

```
bonapanjaitan556@bonapanjaitan556:~$ sudo ufw allow 21  
Rules updated  
Rules updated (v6)  
bonapanjaitan556@bonapanjaitan556:~$ sudo ufw allow 10000:10100/tcp  
Rules updated  
Rules updated (v6)  
bonapanjaitan556@bonapanjaitan556:~$ _
```

10. Kemudian akses dari client dengan IP yang sama adri DHCP buka terminal dan ketik ftp (ip server)



The screenshot shows a terminal window titled "bonapanjaitan556@bonapanjaitan556: ~" with a search bar and window controls. The terminal output shows the user running the command "ftp 10.54.232.71". The output is as follows:

```
bonapanjaitan556@bonapanjaitan556:~$ ftp 10.54.232.71  
Connected to 10.54.232.71.  
220 (vsFTPD 3.0.5)  
Name (10.54.232.71:bonapanjaitan556): userftp  
331 Please specify the password.  
Password:  
500 OOPS: vsftpd: refusing to run with writable root inside chroot()  
ftp: Login failed  
ftp>
```

atau jika ingin menggunakan yang memiliki GUI anda bisa menginstall filezilla di ubuntu client dengan command `sudo apt install filezilla` dan ketika sudah selesai akses filezilla dan masuka IP, username, password, berserta port untuk konek ke FTP server.

Instalasi & Konfigurasi Mail Server pada Ubuntu Server

1. Sebelum install mail server kita perlu memperhatikan bahwa dns bona.local itu punya mail server . MX Record adalah penanda kemana email itu dikirim sehingga kita perlu mengedit file forward dns kita dengan menambahkan MX.

```
GNU nano 7.2 /etc/bind/zones/forward.bona
$TTL 604800
@      IN      SOA      ns1.bona.local. admin.bona.local. (
                        2024052201      ; Serial
                        3600             ; Refresh
                        1800             ; Retry
                        604800           ; Expire
                        86400 )          ; Negative Cache TTL
;
@      IN      NS       ns1.bona.local.
@      IN      A        192.168.12.1
ns1    IN      A        192.168.12.1
server IN      A        192.168.12.1
www    IN      A        192.168.12.1
@      IN      MX       10 mail.bona.local.
mail   IN      A        192.168.12.1
```

2. cek konfigurasi forward nya dengan command `sudo named-checkzone bona.local /etc/bind/zones/forward.bona` dan restart bind9 nya dengan command `sudo systemctl restart bind9`

```
bonapanjaitan556@bonapanjaitan556:~$ sudo named-checkzone bona.local /etc/bind/zones/forward.bona
zone bona.local/IN: loaded serial 2024052201
OK
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl restart bind9
bonapanjaitan556@bonapanjaitan556:~$
```

3. verifikasi mail server dengan command `dig MX Josua.local @192.168.12.1`

```
bonapanjaitan556@bonapanjaitan556:~$ dig MX bona.local @192.168.12.1

; <<> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<> MX bona.local @192.168.12.1
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53128
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 1ae710cd73dfe35a0100000069f503937b0d6bd77b024492 (good)
;; QUESTION SECTION:
;bona.local.                IN      MX

;; ANSWER SECTION:
bona.local.                604800  IN      MX      10 mail.bona.local.

;; ADDITIONAL SECTION:
mail.bona.local.           604800  IN      A        192.168.12.1

;; Query time: 0 msec
;; SERVER: 192.168.12.1#53(192.168.12.1) (UDP)
;; WHEN: Fri May 01 19:48:35 UTC 2026
;; MSG SIZE rcvd: 104

bonapanjaitan556@bonapanjaitan556:~$ _
```

4. Selanjutnya instalasi postfix dengan command `sudo apt install postfix -y` Postfix bertugas menerima dan mengirim email antar server maupun ke mailbox lokal. Gambar dibawah merupakan tampilan sudah berhasil install postfix.

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bonapanjaitan556@bonapanjaitan556:~$ _
```

5. Masuk ke file konfigurasi postfix dengan command `sudo nano /etc/postfix/main.cf` dan mulai lah lakukan konfigurasi , Digambar dibawah saya menghapus semua komentar agar file konfigurasi lebih bersih dan mudah untuk diedit sesuaikan DNS dan network dengan milik kalian sendiri.

```
GNU nano 7.2 /etc/postfix/main.cf *
smtpd_banner = ESMTP
biff = no
append_dot_mydomain = no
myhostname = mail.bona.local
mydomain = bona.local
myorigin =
inet_interfaces = all
inet_protocols = ipv4
mydestination = , localhost, localhost,
mynetworks = 127.0.0.0/8, 192.168.12.0/24
home_mailbox = Maildir/
message_size_limit = 10240000
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_recipient_restrictions =
    permit_mynetworks,
    permit_sasl_authenticated,
    reject_unauth_destination
```

6. Kemudian restart postfix agar konfigurasi kita berjalan

```
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl restart postfix
bonapanjaitan556@bonapanjaitan556:~$
```


7. kemudian install dovecot sudo apt install dovecot-core dovecot-imapd -y yang berfungsi untuk menyimpan email dan melayani client yang ingin membaca email via protokol IMAP.

```
bonapanjaitan556@bonapanjaitan556:~$ sudo apt install dovecot-core dovecot-imapd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libexttextcat-2.0-0 libexttextcat-data
```

8. kemudian buka file konfigurasi dovecot dengan command sudo nano etc/dovecot /dovecot.conf

```
GNU nano 7.2 /etc/dovecot/dovecot.conf
## Dovecot configuration file

# If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration

# "doveconf -n" command gives a clean output of the changed settings. Use it
# instead of copy&pasting files when posting to the Dovecot mailing list.

# '#' character and everything after it is treated as comments. Extra spaces
# and tabs are ignored. If you want to use either of these explicitly, put the
# value inside quotes, eg.: key = "# char and trailing whitespace "

# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.protocol

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, ":::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
listen = *, ::
```

9. konfigurasi Lokasi dari mail box dengan dengan command sudo nano etc/dovecot/conf.d/10-mail.conf

```
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir
```

10. Kemudian konfigurasi file sudo nano /etc/dovecot/conf.d/10-auth.conf konfigurasi ini berkaitan dengan autentikasi user

```
# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also ssl=required setting.
disable_plaintext_auth = no
```

```
# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain login
```

11. kemudian konfigurasi file sudo nano /etc/dovecot/conf.d/10-master.conf Dovecot membuat socket file di lokasi yang bisa dibaca Postfix, sehingga saat user login SMTP, Postfix bisa bertanya ke Dovecot "apakah password ini benar?".

```
GNU nano 7.2 /etc/dovecot/conf.d/10-master.conf
service auth {
  # auth_socket_path points to this userdb socket by default. It's typically
  # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have
  # full permissions to this socket are able to get a list of all usernames and
  # get the results of everyone's userdb lookups.
  #
  # The default 0666 mode allows anyone to connect to the socket, but the
  # userdb lookups will succeed only if the userdb returns an "uid" field that
  # matches the caller process's UID. Also if caller's uid or gid matches the
  # socket's uid or gid the lookup succeeds. Anything else causes a failure.
  #
  # To give the caller full permissions to lookup all users, set the mode to
  # something else than 0666 and Dovecot lets the kernel enforce the
  # permissions (e.g. 0777 allows everyone full permissions).
  unix_listener auth-userdb {
    mode = 0666
    user = dovecot
  }

  # Postfix smtp-auth
  unix_listener /var/spool/postfix/private/auth {
    mode = 0666
    user = postfix
    group = postfix
  }

  # Auth process is run as this user.
  #user = $default_internal_user
}
```

12. restart dovecot agar semua file konfigurasi kita berjalan

```
bonapanjaitan556@bonapanjaitan556:~$ sudo systemctl restart dovecot
bonapanjaitan556@bonapanjaitan556:~$
bonapanjaitan556@bonapanjaitan556:~$ _
```

13. kemudian tambahkan 2 user untuk melakukan pengiriman mail dengan command sudo adduser (namauser) dan otomatis menambahkan password

```

bonapanjaitan556@bonapanjaitan556:~$ sudo adduser bona
info: Adding user `bona' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `bona' (1002) ...
info: Adding new user `bona' (1002) with group `bona (1002)' ...
info: Creating home directory `/home/bona' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bona
Enter the new value, or press ENTER for the default
    Full Name []: bona
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `bona' to supplemental / extra groups `users' ...
info: Adding user `bona' to group `users' ...
bonapanjaitan556@bonapanjaitan556:~$ sudo adduser admin
info: Adding user `admin' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `admin' (1003) ...
info: Adding new user `admin' (1003) with group `admin (1003)' ...
info: Creating home directory `/home/admin' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin
Enter the new value, or press ENTER for the default
    Full Name []: admin
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `admin' to supplemental / extra groups `users' ...
info: Adding user `admin' to group `users' ...
bonapanjaitan556@bonapanjaitan556:~$ _

```

14. kemudian anda bisa mulai mengirim pesan dari user yang anda tambahkan disini saya membuat user bona dan admin

```

bonapanjaitan556@bonapanjaitan556:~$ sudo telnet mail.bona.local 25
Trying 192.168.12.1...
Connected to mail.bona.local.
Escape character is '^]'.
220 ESMTTP
mail from: bona@bona.local
250 2.1.0 Ok
rcpt to: admin@bona.local
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Subject: Test Mail

Selamat Kuliah
.
250 2.0.0 Ok: queued as B6D7F603D8
quit
221 2.0.0 Bye
Connection closed by foreign host.
bonapanjaitan556@bonapanjaitan556:~$

```

15. kemudian kita bisa lihat mail bona disisi admin

```
bonapanjaitan556@bonapanjaitan556:~$ sudo telnet mail.bona.local 110
Trying 192.168.12.1...
Connected to mail.bona.local.
Escape character is '^]'.
+OK Dovecot (Ubuntu) ready.
user admin
+OK
pass bantusubscribe556
+OK Logged in.
stat
+OK 1 439
retr 1
+OK 439 octets
Return-Path: <bona@bona.local>
X-Original-To: admin@bona.local
Delivered-To: admin@bona.local
Received: from bona.local (ns1.bona.local [192.168.12.1])
        by mail.bona.local (Postfix) with ESMTP id 7648C603D8
        for <admin@bona.local>; Fri,  1 May 2026 20:45:35 +0000 (UTC)
Subject: Test Mail
Message-Id: <20260501204543.7648C603D8@mail.bona.local>
Date: Fri,  1 May 2026 20:45:35 +0000 (UTC)
From: bona@bona.local

Selamat Kuliah
.
quit
+OK Logging out.
Connection closed by foreign host.
bonapanjaitan556@bonapanjaitan556:~$
```

SELESSAI.